A New Strategy for Fostering Engineering Students' Entrepreneurial Skills in the School of Entrepreneurs at the University of Brasília

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Abstract

This article presents the school of entrepreneurs at the University of Brasilia and emphasizes its importance in the academic training of students, especially those majoring in engineering. This school is a program within the Technological Development Support Center, which is responsible for the developing the University's innovation policy and disseminating entrepreneurship. We shed light on this school's pedagogy and learning methods, which can qualify students for the professional world. This study discusses the experience of teaching entrepreneurship through the course "Introduction to Business Activity." This course aims to promote and disseminate entrepreneurial culture by developing basic and emerging skills in areas such as entrepreneurship, creativity, and innovation through the creation of new technologies and the emergence of competitive and innovative businesses in the country for undergraduate students' professional preparation. The course's pedagogy proposes a blended learning model with an innovative approach wherein the student's leadership skills are assessed in classroom activities and through distance learning with the use of various tools such as Mobile App, WhatsApp, YouTube, Facebook, and the Moodle platform. We present a descriptive statistical analysis of the sample of students. The results show that 60% of the total students are enrolled in the 15 engineering programs at UnB, evidencing students' growing interest for entrepreneurship, which is not part of all programs. Nevertheless, the courses are offered as an optative and free module course for 1,200 students per year with an average waiting list of 400 students per semester.

Keywords: Entrepreneurial education. Teaching Innovation and Entrepreneurship. University of Brasília.

1 Introduction

In its 57 years of existence, the University of Brasília (UNB) became aware of the possibility of proposing an innovative academic environment and upgrading the curriculum and pedagogy of its academic programs based on the premise of establishing connections among science, research, and development. Thus, the Technological Development Support Center (TDC) was created in 1986. Educators of the Faculty of Technology (FT), especially the Electrical Engineering (ENE) department, responded to a pressing need to recognize the university's importance as an effective space for designing science via the development of basic and applied research. The demand to solve society's problems, particularly those arising from government and companies, has aroused interest to strengthen the triple helix in establishing the necessary dialogue among these agents that are responsible for the development of the country.

A professional's technical training is only partly responsible for their success and development in the labor market. Several other skills and capabilities are needed to remain competitive in today's global context. Transversal, managerial, moral, and creative psychosocial capabilities are essential to the generation of social value and sustaining a competitive advantage (Swiatkiewicz, 2014; Piscopo & Silveira, 2013). Successful professionals and entrepreneurs have skills such as convincing colleagues to invest their personal and organizational resources in business ventures (Hisrich, Peters, & Shepherd, 2014), resilience (Cruz & Moraes, 2013), and coordinating and leading the collective execution of tasks (Nassif, Bastos, & Chaves, 2014). While these skills and capabilities are complex and necessary to create successful ventures, they are not enough. The reality of business is uncertain and turbulent at the international level and still requires the development of owners' capital to train teams and strategic networks, where one learns to lead and be led at the same time (Collinson & Cook, 2006).

Under the supervision of a research professor, Problem-Based Learning (PBL) provides students with the opportunity to solve real problems in society through group projects and the use of modern technology (Martins, 2014). PBL works with the relevant, current issues of society and firms, i.e., real life issues that have yet to be resolved.

In the opinion of Bender (2012), Project-based Learning (PJBL) may be defined using authentic, realistic projects based on a highly motivating and engaging issue, task, or problem to teach students academic content in the context of cooperative efforts to resolve problems. It has currently become an object of great interest from the perspective learning outcomes obtained through its application. Through this methodology, the student can reflect on their ideas, listen to their colleagues, and be heard by them. They can make choices that directly affect the result. Blumenfeld, Soloway, Marx, Krajcik, Guzdial, & Palincsar (1991), states that PJBL should be developed in groups that work together to achieve a common goal, thus suggesting that the outcome of the process can result in realistic products developed autonomously by students.

Team-based Learning (TBL) can be used as a complementary tool to methodologies such as PJBL and PBL (Parmelee, et al, 2012). TBL emerged in the 1970s when Larry Michaelsen presented an educational strategy for classrooms with many students that sought to create opportunities and obtain the benefits of work in small learning groups consisting of five to seven students (Burguess et al. 2014). According to Michaelsen (1998), the teams should be formed to avoid barriers to the group's cohesion. Prior preparation with readings and activities are essential to the process.

In addition to introducing UNB's School of Entrepreneurs, this text aims to discuss the motivation of students who seek more appropriate training in entrepreneurship. We present an analysis of the research we conducted with 1,800 freshmen students in the Introduction to Business Activity (IBA) course on the first day of classes in the last three academic semesters. The results show that 60% of the total students are enrolled in the 15 engineering programs at UNB. Moreover, this study has three other parts. Section two presents the UNB School of Entrepreneurs, and section three discusses the course we examined for this study. Finally, we reveal our conclusions and final considerations.

2 The UNB School of Entrepreneurs

The extension program called Escola Empreend offers undergraduate and extension courses and is an educational environment that favors the relationship among education, research, and extension by using active teaching methodologies to develop entrepreneurial capabilities. The school's goals are to "develop skills in the field of Technological Innovation Management and Entrepreneurship. To execute projects focused on developing entrepreneurial capabilities in the Federal District and other regions. To coordinate extension, specialization, and training courses. To encourage and support young entrepreneurs and university students with professional/entrepreneurial ambitions" (CDT, 2016). Table 1 shows the courses of the UNB School of Entrepreneurs offered every semester to undergraduate students at UNB.

Table 1. Courses Offered by FT and Administered by Empreend/TDC/UNB

Course	Period Offered	Course Type	Number of Students per Course
Introduction to Business Activity	Semester	Blended learning	600
Entrepreneurship and Technological Innovation	Annual	Classroom-based	30
Junior Enterprise 1	Semester	Classroom-based	50
Junior Enterprise 2	Semester	Classroom-based	50
Research in Entrepreneurship and Innovation	Annual	Blended learning	30
Social Technology and Innovation	Semester	Classroom-based	50
Innovation Process and Product Development	Semester	Classroom-based	30

Table 2. Courses Offered by FT and Administered by Empreend/TDC/UNB

Course Name	Description	Methodology	Evaluation
Introduction to Business Activity (IBA)	To promote and disseminate entrepreneurial culture, develop basic and emerging skills in entrepreneurship, creativity, and innovation.	Blended learning with six classroom meetings and nine online meetings using the Moodle platform	Business Plan
Research in Entrepreneurship and Innovation (PEI)	Scientific method. Scientific research. Research project. Literature review. Data analysis. Preparation of scientific articles on entrepreneurship and innovation.	Action-reflection-action in classroom spaces, group dynamics, directed studies, participation in field research, and presentation of results.	Field research and presentation of a scientific article at TDC Business Fair.
Social Technology and Innovation (STI)	Technology, society, and humanity's future. Historical, theoretical, and methodological aspects of social technology. Case studies and qualifications of social technologies in Brazil.	Action Research	Development or qualification of social technologies to resolve social demands in the Federal district and surrounding area.
Innovation Process and Product Development (IPPD)	Position the innovation and product development processes within the design of production systems.	PBL and TBL	Student groups based on the PBL and PJBL methodology.
Junior Enterprise 1 (JE1)	Theoretical and practical knowledge based on three axes – entrepreneurship, innovation, and management.	TBL	Field research
Junior Enterprise 2 (JE 2)	Further theoretical and practical knowledge on topics related to organizational management, enabling student members of the Junior Enterprises to share knowledge.	TBL	Field research
Entrepreneurship and Technological Innovation (ETI)	Add value to the business ideas previously developed in IBA and further the knowledge needed to open a new company.	PjBL	Standard Sebrae (Brazilian Micro and Small Business Support Service) business plan

The point of convergence among them all is the teaching methodology with innovative didactics based on active learning principles, which makes both the educator and the student equally responsible for producing science to solve concrete, real problems. Table 2 presented the courses' descriptions, methodologies, and final evaluation. Every semester, the Emprend offers seven blended learning courses with a practical and theoretical approach to disseminate the culture and knowledge related to entrepreneurship and innovation the purpose is to transform ideas into business and provide academic experience that professionally qualifies the student's training.

3 Success Story: IBA

Introduction to Business Activity (IBA) has a course load of 60 hours per semester with four hours a week. There are no prerequisites, and it is offered to students of all undergraduate majors at the university. The university's system, WEB enrollment, executes course enrollment. IBA has been offered as a blended learning course with six meetings – five of which are held in the TDC building and one held by each group's choice as evidence of the meeting, students are organized into groups of up to five members and they must record a video of up to one minute, upload it to their own YouTube channel, and include the link in an e-mail so that the Distance Education team (EaD) may add the link to the course's playlist. To fulfill the course's 15 mandatory meetings,

there are still nine online meetings. The course's content approaches the following topics: Characteristics of the skills and entrepreneurial profile; creativity and innovation; development of ideas; fundamentals for creating a startup; technical design thinking; ideas and opportunities; business models; concept, structure, preparation steps, and business plan model; business strategy; operational plan; SWOT analysis; Canvas Model; concept and importance of marketing for a successful business venture; marketing strategy, competitive advantage, and the marketing compound; sales price strategy; marketing plan within the context of the business plan; financial planning of the business plan; sales price formation; cash flow setup.

3.1 Classroom Dynamics

The first meeting is the inaugural class held in the auditorium of the TDC. There are 6 classes with 100 students, motivational lecture and/or stories from entrepreneurs as opening event, special guests, preferably graduate students from the University, who are aligned with the subject of course and currently innovative entrepreneurs in the city of Brasilia. Still in the first meeting, we present to the students the intention of the course and the importance of student protagonist, the role of the teacher as advisor of the business ideas presented by the working groups. It is suggested that, for the second meeting, students conduct a research about their area of interest in developing a business.

Between the first and second meetings, students fill out an online form with information about their course, period, age, and interest in business. This information enables the division of the class into groups of 25 students, which become part of a WG (work group) for the second meeting. In order to divide the 25-student groups, we used a scrambling technique, in view of the heterogeneity of the classes in relation to the origin of the courses. By this approach, in the end we managed to form groups that were 100% heterogeneous considering areas and courses in some classes.

The second meeting with the duration of one hour for each WG of 25 students, is called the "market of ideas" where the group presents their business ideas. This meeting is held in a special "U" shaped room for the purpose of creating a business meeting ambience. Students present their ideas and they can be "bought" by each other. The result of this meeting is the composition of five groups with five students each that will mold the idea into a business. This new work group will act together until the end of the course.

In the third meeting, groups of 5 students meet in one of the 3 meeting rooms of the TDC entrance hall called: enterprise room, innovate and research. These rooms have an average capacity of 10 people. The dynamic is to let students debate for up to 40 minutes on the proposal of the work, and the teachers listen to the proposals and guide the discussion. This meeting has early scheduling. In the fourth meeting the groups of five are outside the building and the supervision of the teacher of the class. At this stage, they gather to finalize the elaboration of the Canvas model.

In the fifth meeting:

- The groups of 5 students meet for the last time in individual meeting rooms for guidance with teachers responsible for the classes: enterprise, innovate or research. Then, students prepare for the finalization of the business plan and the planning of a presentation on the 6th meeting. The sixth meeting: Along with the business and Innovation Fair of the TDC, the exhibition of 120 projects in poster format is held in sessions divided by classes.
- All works are evaluated by a board of examiners with 3 external guests with expertise in the market. The 20 best placed on the poster exhibition are invited to the pitch session with oral presentation to another board of examiners but now with investors and startup accelerators who may be interested in supporting the continuity of the ideas proposed. It is no longer used as a measure of performance for the course, but as a strategy of visibility and valorization for offering different awards to the tree best placed groups.

Along of the school semester, the groups of work may, whenever they find it necessary, schedule meetings with the course teacher or just schedule the use of meeting rooms for the group. Another differential of the course is that the team of teachers is divided into pairs to meet each class of the six offered every semester. This team is then responsible for mentoring during the classroom meetings as well as the correction and feedback of the work delivered on the AVA platform. Another task team consisting of two professors and two auxiliaries take care of the demands of distance learning beyond the planning of the face-to-face meetings.

3.2 The Learning Methodology

The pedagogical proposal has an innovative approach, approaching education and digital technology, using social network, social media, their own mobile application, evaluation by means of electronic games and their own interactive game room, environment of Virtual Learning (AVA) and instant message exchange application. The following tools are used: Modern tools, applications and concepts such as gamification, agile methods and active methods such as: PJBL (Project Based Learning), PBL (Problem Based Learning) and TBL (Team Based learning).

Concomitantly to the face-to-face meetings there are activities on the Moodle platform with texts, video lessons and online games on their own platform previously developed and used globally, for example, the ITAEWEB[3] game and the interactive games room. They were developed and maintained by the Laboratory of technological innovations for experience environments. We extend collaborative learning to create a conceptual framework that unite TIC tools, enabling the most accurate analysis to estimate educational benefits in order to improve the choice of learning activities. Not only for courses related to entrepreneurship, but also for others that are grouped in this theoretical-practical perspective. WhatsApp groups are used for every 2 classes, totaling 200 students in each so that students can remedy doubts faster since all students come from in person courses and may have trouble in attending a course in the semi-presential model.

One of the success points of the offer of this course has been the opportunity of dynamism in alternating the dynamics of classes as well as the physical structure of the rooms in which we receive the classes for the classroom meetings. The logistics and physical infrastructure of the TDC building located within the UNB contains an auditorium for 100 people, 1 U-shaped room for deliberative council meeting, 1 games room with 2 interactive tables, 3 individual meeting rooms with capacity for 10 people and 1 room with support for videoconferencing with capacity for up to 30 people, entrance hall with capacity for up to 100 people standing has been used in the pilot project as "classroom" to host the 5 meetings of the course of IBA focus specific to this topic to better demonstrate one of the efforts of the Empreend.

3.3 The Learning Assessment Process

The course is divided in 4 modules. Each teaching unit has a set of theoretical and practical activities with products to be delivered that will consolidate the modeling of the business, considering entrepreneurial competencies in learning to work together and to exercise creativity, through practical resolutions of social and economic problems. Within the evaluation criteria students face the following challenges: the delivery of the Canvas business model, the presentation of this model to a qualified team with the possibility of improvements and the improvement of the idea set within a business that can be feasible in the market. Each module will have individual and/or group assessment through the formation of a group of until five students, who will complete part from the projects in the course together.

1. Entrepreneurial skills:
    a. Submit product Proposal/Innovative service (Individual);
    b. Answering the mentorship questionnaire (individual);
    c. Answer the Quiz Module 1 – Google Forms (individual);
    d. Answer the Quiz ITAEWEB Module 2 (individual).
    e. Participation in the second in-person meeting
2. Canvas Model:
    a. Upload from the Canvas template to the virtual Environment (aprender.unb.br) (group);
    b. Participation in the 3rd meeting in person
3. Business proposal:
    a. Submit complete business plan according to the template provided (group);
    b. Resend business plan with corrections if necessary (according to teacher corrections) (group);
    c. Participation in the 4th virtual meeting

[3] www.itae.unb.br

4. Exhibition/presentation of the work at the business Fair of the TDC:
    a. Participation in the exhibition/presentation of the work at the business fair from the TDC (individual);
    b. Participation in an in-person lecture/event that encourages entrepreneurship until the last day of the semester.

3.4 The public of the various engineering courses and the motivation for the discipline

In the first in person meeting of the IBA course students respond an online questionnaire named "Student Profile" in the Google Forms platform. The quiz is linked by the mobile application of the course with important questions to assess the goal of the public that is interested in the school of entrepreneurs. Analyzing the responses of 3 consecutive semesters (2017/2, 2018/1 and 2018/2), we had as input 60% of the academic's registrants from the 15 Engineering courses offered by UNB. Figure 1 highlights the percentage interest in the course by engineering course the 1005 respondents of the engineering courses.

It was found that although the discipline is offered to the entire university community of UNB, most of the students interested are from the engineering courses of the Faculty of Technology of the Darcy Ribeiro campus. This is probably due to the characteristics of the courses of engineering majors, which lead students to think about developing products and services. Another factor may be related to the geographical proximity from FT to Building of the TDC/UNB at the university campus Darcy Ribeiro.

Still, we believe that balancing the supply of discipline among university campuses, because each of them has a reference in their knowledge field. This has been used as a differential for assembling the working groups within the course. The methodology of the course believes that the engineering professionals also need as in all other areas to work in multidisciplinary teams. And the course has enabled this intersection among the most diverse areas of knowledge.

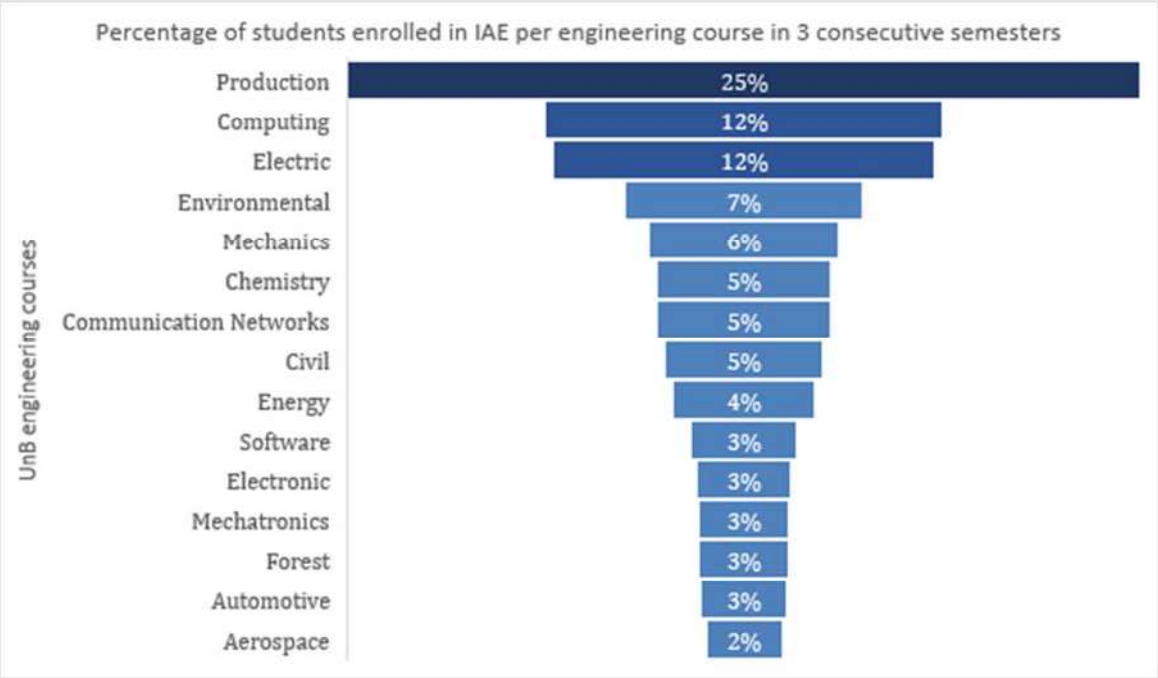

Figure 1. Percentage by engineering course of interest in the discipline - Source: Authors

Figure 2 shows the percentage by type of motivation for the course in view of students interviewed. It is noteworthy that in this sample the discipline is not offered as mandatory discipline, that is, all interested students seek the discipline to compose their curriculums on their own.

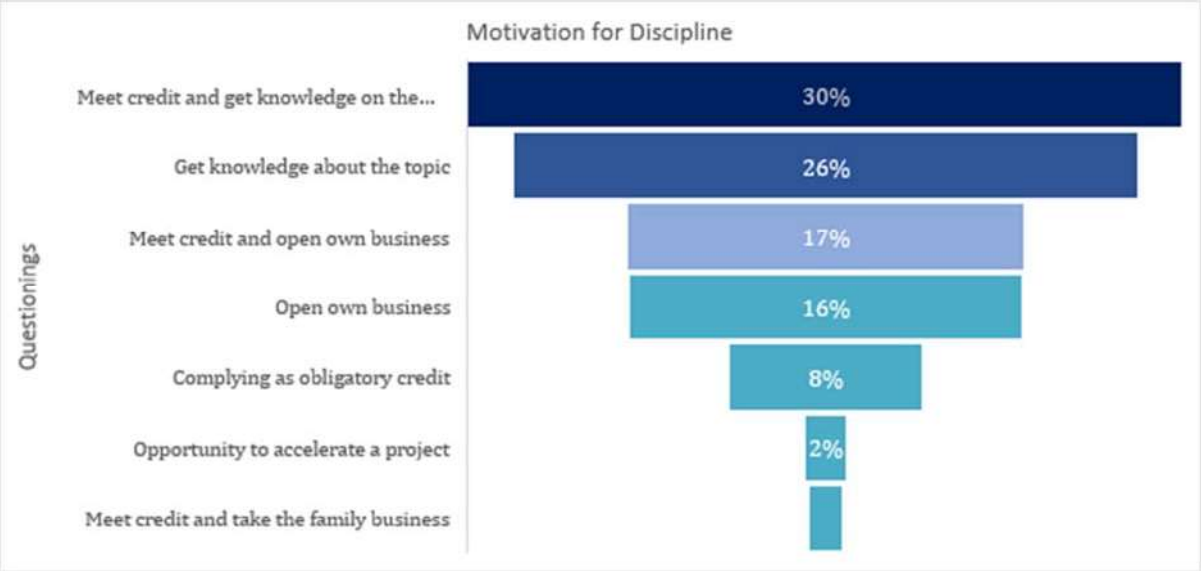

Figure 2. Relation on the motivation type for IBA discipline - Source: Authors

Still in Figure 2, in addition to the first two motivations that suggest the interest in acquiring basic knowledge about the area of course, 56% share this interest. We believe that a possible justification for this high demand, given by the prolonged Brazilian economic crisis with high unemployment rates, entrepreneurship can serve as a professional alternative for an uncertain professional future.

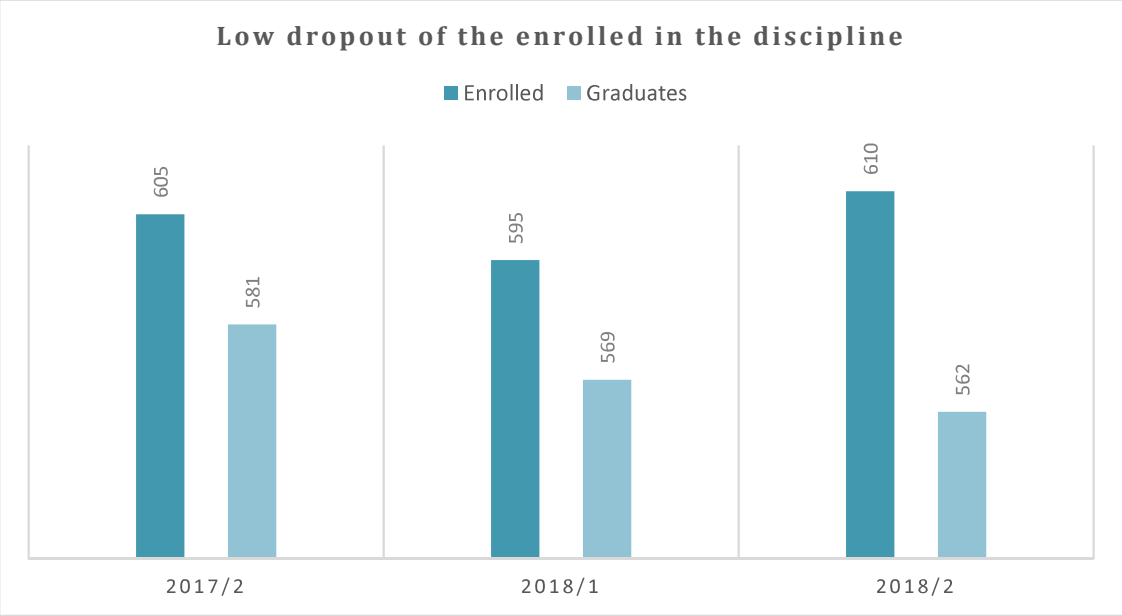

Figure 3. Number of interested in the discipline by semester - Source: Authors

We believe that problem-solving, and learning based on interaction between students are important parts of the process, occupying the longest space of time of the methodology. This set of strategies has contributed to the success of the courses of the Empreend, given the minimum number of evasions in the semester. In the case of IBA, the average utilization of the discipline is 95% as can be demonstrated in Figure 3. Maybe this could explain the average waiting list of 400 students per semester.

4 Conclusion and Future Research

This article presents UNB's School of Entrepreneurs and highlights its importance in the academic training of students, especially engineering students. The school is part of the Technological Development Support Center, which is responsible for developing the university's innovation policy and disseminating entrepreneurship. We draw attention to this school's pedagogy and learning methods, which can qualify students for the professional world. Due to the labor market's new demands, even the most conservative Brazilian educational institutions have shown signs of flexibility regarding their traditional teaching methods. At the same time, the production sector encounters difficulties to recruit skilled workers to work on the frontier of knowledge who have mastered skills and attitudes such as the ability to lead, work in groups, plan and manage strategically, learn independently – the so-called soft skills (Abenge, 2018).

In other words, solid technical training demand is rising, combined with a more humanistic and entrepreneurial formation. At the University of Brasilia, the use of innovative pedagogies with active learning has driven students to think their own business and experience other forms of integration into the world of work. Another aspect is the opportunity to practice entrepreneurial skills by learning to transform ideas into business through the creation of business plans. But the important thing has been learning to work as a team and thinking solutions for the improvement of life in society.

The course contributes to the promotion of the innovation culture and entrepreneurship at UNB and reached, in 2017, the first place among the higher education entrepreneurial institutions of the Midwest, according to the ranking of the Brazilian Confederation of Companies Junior. In addition to achieving the best performance in the region, UNB began to occupy the 8th position in the national classification in 2017. In 2016, the UNB was in the 18th position.

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